The examination is being carried out on the **following application documents**

Description, Pages

1-4, 6-11	as originally filed	
5, 5a	filed in electronic form on	21-08-2018

Claims, Numbers

1-12	filed in electronic form on	21-05-2019

Drawings, Sheets

1/11-11/11	as originally filed

1 Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 DASS S C ET AL: "Validating a Biometric Authentication System: Sample Size Requirements",
IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE COMPUTER SOCIETY, USA,
vol. 28, no. 12, 1 December 2006 (2006-12-01), pages 1902-1913, XP001523408,
ISSN: 0162-8828, DOI: 10.1109/TPAMI.2006.255

2 The Applicant filed amendments in electronic form on 21/05/2019 replacing the claims with new claims 1-12.

3 The present application does not meet the requirements of Art.123(2) EPC since the amendments filed with the letter dated 21/05/2019 introduce subject-matter which extends beyond the content of the application as originally filed because the amendments cover embodiments which were not disclosed. In **claims 1**, **6** and **11** it was added the step of "the second estimator is configured to adopt the combined false acceptance rate in a range of the similarity degree threshold, in which at least one of a statistic amount related to variability of false acceptance rates of the plurality of sets

and the combined false acceptance rate is a threshold or less (, is adopted)". In addition (or due to) to the fact that this phrase is not clear (see objection in section 4), the support for this step in the indicated description paragraphs [0036] and [0039] cannot be found.

4 The application does not meet the requirements of Article 84 EPC, for the following reasons:

4.1 **Claims 1**, **6** and **11** are not clear because the phrase: -"the second estimator is configured to adopt the combined false acceptance rate in a range of the similarity degree threshold, in which at least one of a statistic amount related to variability of false acceptance rates of the plurality of sets and the combined false acceptance rate is a threshold or less" from **claim 1** or "the second estimator is configured to adopt the combined false acceptance rate in a range of the similarity degree threshold, in which at least one of a statistic amount related to variability of false acceptance rates of the plurality of sets and the combined false acceptance rate is a threshold or less, is adopted" **from claims 6** and **11**-, is not clear, because:

4.1.1 It is not clear what is adopted and how.

4.1.2 It is not clear whether the "combined false acceptance rate" is adopted or not in the estimation of the "combined false acceptance rate".

4.1.3 It is not clear how "combined false acceptance rate" is adopted in the estimation of the "combined false acceptance rate".

4.1.4 It is not clear whether the "similarity degree threshold" is adopted or not in the estimation of the "combined false acceptance rate".

4.1.5 It is not clear how the "similarity degree threshold" is adopted in the estimation of the "combined false acceptance rate".

4.1.6 The phrase "at least one of a statistic amount" is not clear because it should be followed by an or to make it clear one of what options should be chosen.

4.1.7 Due to the phrase "at least one of a statistic amount" it is not clear whether the statistic amount depend on both the "false acceptance rates of the plurality of sets" and "the combined false acceptance rate" or only one of said rates.

4.1.8 It is not clear what "is" a threshold.

4.1.9 It is not clear whether the value that is "a threshold" is the same value which represents "the similarity degree threshold".

4.1.10 It is not clear how something is either a threshold or less.

4.2 Due to the clarity issues raised in 4.1, the technical effect of the combined features of **claims 1, 6** and **11** is not clear. It is not clear for what is needed to be used the "combined false acceptance rate" and/or what is evaluated based on the "combined false acceptance rate".

5 The present application does not meet the requirements of Art. 52(1) EPC since the subject-matter of independent **claim 1** is not novel within the meaning of Article 54(1) and (2) EPC:

5.1 Document **D1** discloses in accordance with the following features of **claim 1** (the references in parentheses applying to this document):

An evaluation device (Abstract: "Authentication systems based on biometric features") comprising:
a generator configured to generate a plurality of sets (**page 1908, column right, first paragraph, section 5**: "In order to avoid overfitting, ... Instead of selecting all four fingers, we choose only c = 2, namely, the right index and right middle fingers, and use the d = 2 impressions per finger ") from a first population of biometric data collected from a plurality of users (**page 1908, column right, first paragraph, section 5**: "The Michigan State University (MSU) database consists of fingerprint impressions from four different fingers (the right index, right middle, left index, and left middle fingers) of 160 users."), the number of biometric data of the plurality of sets being smaller than that of the first population (**page 1908, column right, first paragraph, section 5**: "In order to avoid overfitting, we reduce "), at least each part of the plurality of sets being different from each other (**page 1904, column right, second paragraph, section 2**: "multiple biometric entities (e.g., different fingers)");
a first estimator configured to estimate a probability density distribution (**formula 11** *for showing the dependency of the common distribution function "F" from "s")* of similarity degrees of different user's pairs (**page 1905, column right, first paragraph, section 2**: "where *s(i, k)* is the generic score corresponding to the *k*th component of *Si*") of each of the plurality of sets (**page 1904, column right, second paragraph, section 2**: "To obtain similarity scores, a pair of biometric samples, [Beta] and [Beta]' with [Beta]' [different from]

[Beta], are taken from B and a matcher S is applied to them, resulting in the similarity score S([Beta], [Beta]' " and **formula 8** *showing dependence of "s" from "S"*);
a calculator configured to calculate a false acceptance rate (**page 1902, column left, first paragraph, section 1**: "A threshold, *[lambda]*, is specified so that all similarity values ... greater than *[lambda]* lead to ... acceptance" and **figure 1a** and **formula 11**, which implicitly mean FAR*k[lambda]*=P{*s(i,k)>[lambda]}=1-*(P{*s(i,k)<[lambda]})*= 1 - F*k[lambda] "*) with respect to each similarity degree threshold on a basis of the probability density distribution (**formula 11** "P{*s(i,k)<[sk]}"*), with respect to each of the plurality of sets; and
a second estimator configured to estimate a combined false acceptance rate from a statistic amount of false acceptance rates of the plurality of sets calculated by the calculator, the combined false acceptance rate being a false acceptance rate of the first population of biometric data. (**formula 18**, "the population FAR" *as a function of FAR for each plurality of sets* " P*(s(i,k)>[lambda]*").
wherein:
the second estimator is configured to adopt the combined false acceptance rate in a range of the similarity degree threshold, in which at least one of a statistic amount related to variability of false acceptance rates of the plurality of sets and the combined false acceptance rate is a threshold or less (**formula 18** *wherein the range of similarity degrees is given by the probability that the similarity degrees are bigger than a threshold [lampda]* and **formula 16** *regarding the relation of similarity degrees to the variability of "FAR".*); and
the statistic amount is a standard deviation, a dispersion, an upper limit or a lower limit of the false acceptance rates (**formula 16** and **18** *disclosing the "false acceptance rate threshold [lampda]" and "the probability I that a value is above a **lower** limit [lampda]" corresponding to the "statistic amount" in **claim 1**. In addition the other possibilities for mapping the "statistic amount" in **claim 1** are obviously derivable from the quantities disclosed in the **formulas 16** and **18**).*

6 Referring to the objection raised above, **claims 6** and **11** are also not allowable for lack of novelty (Art. 54 EPC) since their subject-matter corresponds to that of **claim 1**.

7 Additionally, **all** the dependent **claims** do not appear to contain any additional technical features which, either alone or in combination with the features of any claim to which they refer, render the present subject-matter novel (Art. 54 EPC) or inventive (Art. 56 EPC) since their features are known or derivable from the prior **D1** (see cited passages in the Search Report) alone or in combination with **document D3**.

8 Moreover, it should be noted that the teachings of document **D2** (see cited passages in the Search Report), taken alone, are also relevant to the question of inventive step (Art. 56 EPC) regarding the subject-matter of the present independent **claims 1, 6** and **11**, since **D2** contains (see cited passages in the Search Report) the estimation of the probability density distribution of the similarity degrees, the calculation of the false acceptance rate for both the individual sets and the combination of all sets. The difference with **D2** is that in **D2** the generation of subsets out of the main set is not mentioned. In **D2,** is mentioned that the confidentially level depends on the number of biometric data contained in the set and is reduced if less data are available. Therefore, a skilled person in the art would reduce the value of the number of biometric data in the set of **D2**, to generate subsets which contain less data to have a faster false acceptance rate estimation at the cost of less reliability.

9 In his letter dated 21/05/2019 the applicant indicated that **Documents D1** and **D2** do:

a) not disclose at least amended feature of "the second estimator is configured to adopt the combined false acceptance rate in a range of the similarity degree threshold, in which at least one of a statistic amount related to variability of false acceptance rates of the plurality of sets and the combined false acceptance rate is a threshold or less ".

b) not disclose at least amended feature of the statistic amount is a standard deviation, a dispersion, an upper limit or a lower limit of the false acceptance rates".

9.1 However, the Examining Division is of the opinion that:

regarding point a), said feature is not allowed under Art.123(2) EPC and Art. 84 EPC. In addition, if it would become clear and allowable, it could be disclosed in **Document D1** (see feature mapping in section 5.1, especially

the underlined part).
regarding point b), said feature is disclosed in **Document D1** (see feature mapping in section 5.1, especially the underlined part).

10 In case the Applicant intends to file an amended set of claims the following remarks should also be taken into consideration:

10.1 When filing amended claims the Applicant should at the same time bring the Description into conformity with the amended claims (Rule 42(1)(c) EPC).

10.2 In order to facilitate the examination of the conformity of the amended application with the requirements of Art. 123(2) EPC, the Applicant is requested to clearly identify the amendments carried out and to indicate the passages of the original application as filed on which these amendments are based (Guidelines, H-III, 2.2).